

DR. MARIAM AL KHATIB

Vice President – Satellite Operations (GEO/LEO/MEO Operations | Fleet Health & Availability | Automation & AI)

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Executive Summary

Satellite operations executive with 16+ years of experience in satellite telecommunications operations and mission assurance, spanning GEO fleet station-keeping, LEO/MEO operational readiness, anomaly investigation, ground segment oversight, and enterprise-scale operations governance. Proven leader of multi-disciplinary operations teams (24/7 control centers) with a track record of improving availability KPIs, strengthening contingency response, and driving autonomous operations initiatives to reduce human error. Experienced in strategic planning, budgeting, procurement (SCS/FDS), vendor governance, and executive reporting to CTO-level stakeholders.

Core Leadership & Domain Competencies

- Satellite fleet operations leadership (24/7) • GEO station-keeping & orbital slot management • ITU filings / orbital parameter compliance • Commanding & monitoring, routine & contingency ops • Anomaly investigation & resolution governance • Ground segment oversight (SCS/FDS, TT&C; mission planning) • Operations KPIs (availability, MTTR, command success rate) • Budgeting & cost control • Procurement & RFP leadership • Vendor/contract performance management • Staff development, performance management, succession planning • Automation & AI for autonomous operations • Executive reporting & risk communication

Professional Experience

Director, Satellite Operations (Fleet & Control Centers)

OrbitLink Space Services – Abu Dhabi, UAE | Jan 2019 – Present

- Led end-to-end operations for a mixed satellite communications portfolio, overseeing control center performance, fleet health, and service availability across commercial and government customers.
- Implemented KPI-driven governance (availability, anomaly response times, change success rate), delivering a sustained **+1.8% fleet availability improvement** over 24 months.
- Directed routine and contingency operations, ensuring GEO spacecraft remained within assigned orbital slots and LEO/MEO assets remained within required orbital parameters and regulatory commitments.
- Chaired anomaly review boards and established investigation playbooks; improved time-to-root-cause by standardizing telemetry triage, commanding approvals, and vendor escalation paths.
- Drove autonomous operations roadmap: automated housekeeping checks, scheduling validation, and alert correlation—reducing operator workload and human error incidents by 30% (YoY).
- Owned annual divisional budget planning and cost control; identified efficiency opportunities in vendor contracts and shift operations while maintaining service SLAs.

Senior Manager, Mission Operations & Ground Systems

SkyWave Satellite Telecom – Dubai, UAE | Jun 2014 – Dec 2018

- Managed mission operations teams supporting commanding/monitoring, mission planning, and ground systems coordination across multiple programs.
- Oversaw SCS/FDS lifecycle upgrades and operational acceptance; coordinated technical requirements with legal, finance, and end users to ensure RFP completeness and stakeholder alignment.
- Established contractor performance monitoring and structured knowledge transfer clauses; verified operator training outcomes and ensured internal teams reached independent operating capability.
- Led project schedules and budget tracking for ground segment modernization, ensuring proper closure and operational readiness milestones.

Satellite Operations Engineer → Lead Operations Engineer

MENA Satcom Operator – Abu Dhabi, UAE | Aug 2008 – May 2014

- Executed day-to-day spacecraft commanding and monitoring; conducted station-keeping support, housekeeping, and maintenance planning.
- Supported contingency operations and post-event analysis; produced incident reports and corrective actions for recurring anomalies.
- Contributed to procedure development and shift training, progressing to lead engineer role within the control center operations team.

Selected Strategic Initiatives

- **Autonomous Ops Program (2022–2024):** Introduced automation for telemetry checks, alert triage, and schedule validation; reduced manual checks and improved incident response consistency.
- **SCS/FDS Procurement & Upgrade (2017–2018):** Led cross-functional RFP process (technical + legal + commercial), vendor evaluation, and implementation governance; ensured knowledge transfer and operational handover.
- **Operational Excellence & Reporting (2019–2021):** Standardized management reporting on fleet health, anomalies, and operational risk; improved executive visibility and decision turnaround time.

Education

Master of Science (MSc), Aerospace Engineering

Khalifa University – Abu Dhabi, UAE | 2006 – 2008

Bachelor of Engineering (BEng), Electrical & Electronic Engineering

University of Sharjah – Sharjah, UAE | 2002 – 2006

Professional Certifications

- Certified Satellite Operations Professional (CSOP) – International Space Operations Association (ISOA)
- Project Management Professional (PMP)
- ITIL Foundation (Service Management)
- Spacecraft Anomaly Investigation & Mission Assurance (specialized training)

Technical & Operational Expertise

Operations: Commanding/monitoring, routine & contingency ops, anomaly management, procedure definition, ops readiness reviews

Orbit & Fleet Compliance: GEO slot management, station-keeping coordination, ITU/orbital parameter compliance (as applicable)

Ground Segment: Spacecraft Control Systems (SCS), Flight Dynamics Systems (FDS), mission planning, TT&C; interfaces (ops oversight)

Automation & AI: Alert correlation, autonomous scheduling validation, rule-based + ML-assisted triage concepts, human-in-the-loop governance

Executive Management: Budget planning, cost control, vendor governance, risk reporting, stakeholder communication to CTO-level

Leadership & People Development

- Led multi-team operations organizations including shift operations, mission planning, and ground systems support (30–60 staff).
- Performance management: objective setting, coaching, formal feedback, and annual appraisals; built succession plans for key operational roles.
- Designed training pathways and competency matrices to prepare engineers for lead roles and improve operational resilience.

Spacecraft Insurance & Claims Support

Supported insurance discussions by consolidating operational evidence, anomaly timelines, and vendor findings to maximize claim outcomes where applicable; ensured documentation met contractual and audit expectations.

Languages

Arabic (Native) • English (Fluent) • French (Basic)

Interests

Space mission assurance • Operational automation • Leadership mentoring • STEM outreach in UAE